Differentially Private Stochastic Convex Optimization under a Quantile Loss Function

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Abstract

We study (ε, δ)-differentially private (DP) stochastic convex optimization under an r-quantile loss function taking the form $c(u) = ru^+ + (1-r)(-u)^+$. The function is nonsmooth, and we propose to approximate it with a smooth function obtained by convolution smoothing technique. The approximated smooth function is suitable for many standard DP algorithms, such as gradient perturbation and objective perturbation, and working on it does not introduce additional computation burden (compared to its non-private counterparts). We first apply gradient perturbation and show that it achieves the optimal excess population risk at the rate $\tilde{O}(\max\{\frac{1}{\sqrt{n}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\})$. We further apply objective perturbation and derive an upper bound on the parameter estimation error $\tilde{O}(\max\{\sqrt{\frac{d}{n}}, \sqrt{\frac{d\ln(1/\delta)}{n\varepsilon}}\})$ under mild assumptions on data generating processes. Some applications in quantile regression and inventory control will be discussed.

1. Introduction

Notation: We use $\mathcal{X} \times \mathcal{Y} \subseteq \mathcal{B}(B_x) \times \mathbb{R}$ to denote the data domain, where $\mathcal{B}(B)$ is a Euclidean ball with radius B. The quantile loss function is $c(u) = ru +^+ (1-r)(-u)^+, \forall c \in \mathbb{R}$, and we let $\ell(\boldsymbol{\theta}; \boldsymbol{x}, y) := c(y - \boldsymbol{\theta}^\top \boldsymbol{x})$, a loss function that takes a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$ and a data point $(\boldsymbol{x}, y) \in \mathcal{X} \times \mathcal{Y}$ as inputs and outputs a real value. The empirical risk of $\boldsymbol{\theta} \in \mathbb{R}^d$ w.r.t. loss ℓ and dataset $\mathcal{D} := \{\boldsymbol{x}_i, y_i\}_{i=1}^n$ is defined as $\widehat{\mathcal{L}}(\boldsymbol{\theta}; \mathcal{D}) := \frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$. The generalization risk of $\boldsymbol{\theta}$ w.r.t. loss ℓ and distribution $\mathbb{P}$ is defined as $\mathcal{L}(\boldsymbol{\theta}; \mathbb{P}) := \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}[\ell(\boldsymbol{\theta}; \boldsymbol{x}, y)]$. Shorthand $\widehat{\mathcal{L}}(\boldsymbol{\theta})$ and $\mathcal{L}(\boldsymbol{\theta})$ are used for the empirical and generalization risk when the context is clear.

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2. Preliminaries

Definition 2.1 (Differential privacy)**.** A randomized algorithm $\mathcal{M} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{Z}$ is (ε, δ)-differential private if, for any pair of neighboring datasets $\mathcal{D} \sim \mathcal{D}'$ that differ in one data point, and for any subset $\mathcal{S} \subseteq \mathcal{Z}$, we have

$$\Pr[\mathcal{M}(\mathcal{D}) \in \mathcal{S}] \leq e^\varepsilon \cdot \Pr[\mathcal{M}(\mathcal{D}') \in \mathcal{S}] + \delta.$$

Definition 2.2 (L-Lipschitz continuity)**.** Let $L > 0$. A function $\ell : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz with respect to norm $\|\cdot\|_2$ over a set $\mathcal{B}$ if for every $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{B}$, we have

$$|\ell(\boldsymbol{\theta}_1) - \ell(\boldsymbol{\theta}_2)| \leq L \cdot \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 .$$

Definition 2.3 (α-strongly convexity)**.** Let $\alpha > 0$. A function $\ell : \mathbb{R}^d \to \mathbb{R}$ is α-strongly convex over a set $\mathcal{B}$ if for every $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{B}$, we have

$$\ell(\boldsymbol{\theta}_1) \geq \ell(\boldsymbol{\theta}_2) + \langle \nabla \ell(\boldsymbol{\theta}_2), \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2 \rangle + \frac{\alpha}{2}\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2^2 .$$

Definition 2.4 (β-smoothness)**.** Let $\beta > 0$. A function $\ell : \mathbb{R}^d \to \mathbb{R}$ is β-smooth over a set $\mathcal{B}$ if for every $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{B}$, we have

$$\ell(\boldsymbol{\theta}_1) \leq \ell(\boldsymbol{\theta}_2) + \langle \nabla \ell(\boldsymbol{\theta}_2), \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2 \rangle + \frac{\beta}{2}\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2^2 .$$

Definition 2.5 (τ-uniform stability)**.** A randomized algorithm $\widehat{\boldsymbol{\theta}} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathbb{R}^d$ is τ-uniform stable with respect to function $\ell : \mathbb{R}^d \times \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ if for any pair of neighboring datasets $\mathcal{D} \sim \mathcal{D}'$, we have

$$\sup_{\boldsymbol{x},y} \mathbb{E}_{\widehat{\boldsymbol{\theta}}}\left[\ell(\widehat{\boldsymbol{\theta}}(\mathcal{D}); \boldsymbol{x}, y) - \ell(\widehat{\boldsymbol{\theta}}(\mathcal{D}'); \boldsymbol{x}, y)\right] \leq \tau,$$

where $\mathcal{D} \sim \mathcal{D}'$ are neighboring datasets differ in one data point only.

Lemma 2.6. *(Bousquet & Elisseeff, 2002) Let $\widehat{\boldsymbol{\theta}} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathbb{R}^d$ be a τ-uniform stable algorithm w.r.t. loss function $\ell : \mathbb{R}^d \times \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. Let $\mathbb{P}$ be a distribution over $\mathcal{X} \times \mathcal{Y}$, and $\mathcal{D} \sim \mathbb{P}^n$ be samples i.i.d. drawn from $\mathbb{P}$. Then, we have*

$$\mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n, \widehat{\boldsymbol{\theta}}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}(\mathcal{D}); \mathbb{P}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}(\mathcal{D}); \mathcal{D})\right] \leq \tau.$$

We are particularly interested in bounding the excess generalization risk of a given differentially private policy π:

$$\mathcal{R}(\pi;\mathbb{P}) := \mathbb{E}_{\mathcal{D}\sim\mathbb{P}^n,\pi}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\pi})\right] - \mathcal{L}(\boldsymbol{\theta}^*),$$

where $\mathcal{L}(\boldsymbol{\theta}) := \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[\ell(\boldsymbol{\theta};\boldsymbol{x},y)\right]$ is the expected loss of a given vector $\boldsymbol{\theta}$, and $\boldsymbol{\theta}^*$ is the optimal vector that an oracle with full information of $\mathbb{P}$ can achieve. Following literature in learning theory, the excess generalization risk can be decomposed into two terms,

$$\begin{aligned}\mathcal{R}(\pi;\mathbb{P}) &= \mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\pi}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}^{\pi})\right] + \mathbb{E}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}^{\pi}) - \mathcal{L}(\boldsymbol{\theta}^*)\right] \\ &= \mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}^{\pi}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}^{\pi})\right] + \mathbb{E}\left[\widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}^{\pi}) - \widehat{\mathcal{L}}(\boldsymbol{\theta}^*)\right],\end{aligned} \tag{1}$$

where all expectations are taken over $\mathcal{D}$ and π, and the second equality comes from $\mathcal{L}(\boldsymbol{\theta}^*) = \mathbb{E}_{\mathbb{P}^n}\left[\widehat{\mathcal{L}}(\boldsymbol{\theta}^*)\right]$. The above risk decomposition essentially follows a framework in learning theory that, loosely speaking, uniform stability plus shrinking ERM imply learnability. Indeed, the first term can be controlled by Lemma 2.6, and the second term corresponds to the excess empirical risk. Our risk analysis will heavily rely on (1).

3. Private Algorithms with Convolution Smoothing

In this section, we first show how convolution smoothing works and its impact. Specifically, we exploit the special structure of the quantile loss function, and derive properties of the smoothed loss function. Then, stochastic gradient perturbation and objective perturbation policies are applied, along with privacy analysis and excess generalization risk analysis.

3.1. Convolution Smoothing

Convolution smoothing is a powerful tool to approximate a non-smooth function with a smooth function generated from convolution between the original function and a properly chosen Kernel Function (also known as Approximate Identity). The idea is also used for Kernel Density Estimation and Fourier analysis.

Condition 1 (Kernel Functions). Let $K : \mathbb{R} \to \mathbb{R}_+$ be a nonnegative function having following properties

- **Integrate to 1**: $\int K = 1$;
- **Symmetry**: $K(u) = K(-u)$ for all $u \in \mathbb{R}$;
- **Monotonicity**: $K(u) \le K(v)$, $\forall |u| \ge |v| \in \mathbb{R}$.
- **Finite moments**: $\kappa_1 := \int_{-\infty}^{\infty} |u|\, K(u)\, du < \infty$, $\kappa_2 := \int_{-\infty}^{\infty} u^2 K(u)\, du < \infty$, and $\overline{K} := \sup_u K(u) < \infty$.

Any function satisfies Condition 1 is suitable for our problems. Commonly used kernels are Gaussian Kernel, Logistic Kernel, Uniform Kernel, and Epanechnikov Kernel, summarized in Table 1 under column "$K(u)$".

Given a bandwidth $h > 0$, and a kernel function K, we follow notations in literature and denote, for any $u \in \mathbb{R}$, $K_h(u) := K(u/h)/h$, $\mathcal{K}(u) := \int_{-\infty}^{u} K(v)\, dv$, $\mathcal{K}_h(u) := \mathcal{K}(u/h)$. Then, against the quantile loss functon $c(u) = ru^+ + (1-r)(-u)^+ = |u|/2 + (r-1/2)u$, the smoothed loss function obtained via convolution smoothing is

$$\begin{aligned}c_h(u) &:= (c * K_h)(u) \\ &= \int_{-\infty}^{\infty} c(v) K_h(u-v)\, dv \\ &= \frac{h}{2}\int_{-\infty}^{\infty} \left|\frac{u}{h} + v\right| K(v)\, dv + \left(r - \frac{1}{2}\right) u.\end{aligned} \tag{2}$$

With previously mentioned kernel functions, the integral of the first term in (2) has closed-form as shown under the third column in Table 1. To illustrate the impact of convolution smoothing, we visualize c_h with Gaussian kernel function in Figure 1. Mathematically, it can be shown that $c_h(\cdot)$ is

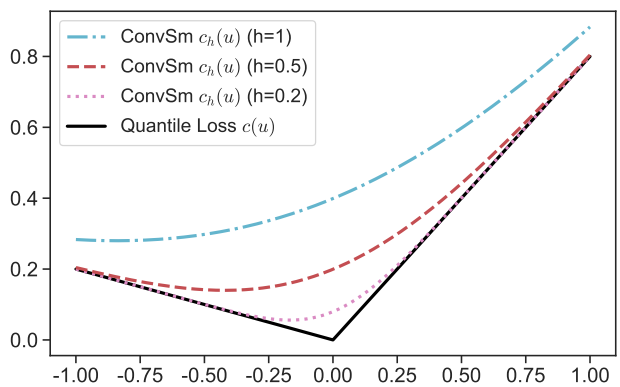

Figure 1. Convolution smoothing with Gaussian kernel. $r = 0.8$

convex, epi-graphically converges to $c(\cdot)$ as $h \to 0$, and second-order differentiable:

$$c_h'(u) = \mathcal{K}_h(u) + r - 1; \quad c_h''(u) = K_h(u) \ \ge 0.$$

These properties are graphically confirmed by Figure 1.

Similarly, the regression loss function $\ell_h(\boldsymbol{\theta};\boldsymbol{x},y) := c_h(y - \boldsymbol{\theta}^\top \boldsymbol{x})$ posseses same properties, and we provide a summary below for later reference.

Lemma 3.1 (Properties of ℓ_h)**.** *Suppose kernel function K satisfies Condition 1. And define $\ell_h(\boldsymbol{\theta};\boldsymbol{x},y) := c_h(y - \boldsymbol{\theta}^\top \boldsymbol{x})$, then function $\ell_h : \mathbb{R}^d \times \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ satisfies following properties. For any $\boldsymbol{\theta}, \boldsymbol{x}, y$,*

Table 1. Kernel Functions

Kernels	$K(u)$	$\int_{-\infty}^{\infty} \lvert u/h + v\rvert\, K(v)\, dv$	$\mathcal{K}_h(u)$
Gaussian	$\frac{1}{\sqrt{2\pi}} e^{\frac{-u^2}{2}}$	$\sqrt{\frac{2}{\pi}} e^{-\frac{u^2}{2}} + u(1-2\Phi(-u))$	$\Phi(u/h)$
Logistic	$\frac{e^{-u}}{(1+e^{-u})^2}$	$u + 2\ln(1+e^{-u})$	$(1+e^{-u/h})^{-1}$
Uniform	$\frac{\mathbb{1}\{\lvert u\rvert \le 1\}}{2}$	$\begin{cases} \frac{u^2}{2} + \frac{1}{2}, \text{ if } \lvert u\rvert \le 1 \\ \lvert u\rvert, \text{ o.w.} \end{cases}$	$\min\{(u/h+1)/2, 1\}\mathbb{1}\{u/h \ge -1\}$
Epanechnikov	$\frac{3}{4}(1-u^2)\mathbb{1}\{\lvert u\rvert \le 1\}$	$\begin{cases} \frac{3u^2}{4} - \frac{u^4}{8} + \frac{3}{8}, \text{ if } \lvert u\rvert \le 1 \\ \lvert u\rvert, \text{ o.w.} \end{cases}$	$\begin{cases} -\frac{u^3}{4h^3} + \frac{3u}{4h} + \frac{1}{2}, \text{ if } \lvert u/h\rvert \le 1 \\ \mathbb{1}\{u/h \ge 1\}, \text{ o.w.} \end{cases}$

1. *function ℓ_h is second-order differentiable in $\boldsymbol{\theta}$ with*

$$\nabla \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) = \left[\mathcal{K}_h\left(\boldsymbol{\theta}^\top \boldsymbol{x} - y\right) - r\right] \cdot \boldsymbol{x};$$
$$\nabla^2 \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) = K_h\left(y - \boldsymbol{\theta}^\top \boldsymbol{x}\right) \cdot \boldsymbol{x}\boldsymbol{x}^\top \succcurlyeq \mathbf{0};$$

2. *function ℓ_h is $\bar{r}B_x =:$ L-Lipschitz continuous and $\overline{K}B_x^2/h =: \beta$-smooth in $\boldsymbol{\theta}$ with respect to $\|\cdot\|_2$;*

3. *function ℓ_h is upper and lower bounded by affine functions of ℓ,*

$$\ell(\boldsymbol{\theta}; \boldsymbol{x}, y) \le \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) \le \ell(\boldsymbol{\theta}; \boldsymbol{x}, y) + \frac{1}{2} h \kappa_1.$$

Computational Efficiency: We highlight that gradients $\nabla \ell_h$ (or Hessian matrix $\nabla^2 \ell_h$) can be efficiently computed: one can simply replace $\mathcal{K}_h$ (or K_h) in Lemma 3.1 part 1 with the corresponding expression under fourth (or second) column in Table 1. Such a computational advantage is not observed in other approaches with proximal operators to addressing nonsmoothness, such as Moreau envelope (Bassily et al., 2019), a localization-based Phased-ERM (Feldman et al., 2020), and noisy Mirror Descent (Asi et al., 2021), where one must solve a convex optimization for the gradient.

With Lemma 3.1, the excess generalization risk (1) of a given vector $\widehat{\boldsymbol{\theta}}_h^\pi$ under policy π_h with convolution smoothing bandwidth $h > 0$ can be further upper bounded:

$$\mathcal{R}(\pi_h; \mathbb{P}) \le \mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}^\pi) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}^\pi)\right] + \mathbb{E}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^\pi) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] + \frac{1}{2} h \kappa_1,$$

where the inequality comes from properties in part 3 of Lemma 3.1, since $\widehat{\mathcal{L}}$ and $\mathcal{L}$ are merely linear operations over ℓ, which preserves the properties. The above analysis implies that convolution smoothing will introduce an additional linear term in bandwidth $h > 0$ and kernel function's statistics κ_1. Thus, if our polices are uniformly stable, ensure shrinking ERM and a vanishing bandwidth h, the excess generalization risk will converge to 0. We next show that some standard differentially private algorithms are eligible.

3.2. Stochastic Gradient Perturbation Policy

Stochastic Gradient Perturbation (SGP) policy is based on standard gradient descent by injecting a well-calibrated Gaussian noise into the gradient in each iteration. We apply SGP to smoothed quantile loss function ℓ_h rather than ℓ directly. The nuance appears in Step 5 of Algorithm SGP_h formally given below. We now state privacy and excess generalization risk guarantees of Algorithm SGP_h.

Algorithm 1 Stochastic Gradient Perturbation (SGP_h)

Input: Private dataset $\mathcal{D}$, privacy parameters $\varepsilon \le 1$, $\delta \ge 0$, kernel function K with bandwidth $h > 0$, Lipschitz parameter $L = \bar{r}B_x$, smoothness parameter $\beta = \overline{K}B_x^2/h$, noise variance $\sigma^2 = 8L^2 \ln(1/\delta)/\varepsilon^2$, step size $\eta > 0$.

1: Set initial point $\widehat{\boldsymbol{\theta}}_{h,1} = \mathbf{0}$
2: **for** $t = 1$ to $n^2 - 1$ **do**
3: Uniformly sample a record $(\boldsymbol{x}_{(t)}, y_{(t)})$ from $\mathcal{D}$ with replacement
4: Sample a noise vector $\boldsymbol{w}_t \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{d\times d})$
5: $\widehat{\boldsymbol{\theta}}_{h,t+1} \leftarrow \widehat{\boldsymbol{\theta}}_{h,t} - \eta \cdot (\nabla \ell_h(\widehat{\boldsymbol{\theta}}_{h,t}; \boldsymbol{x}_{(t)}, y_{(t)}) + \boldsymbol{w}_t)$
6: **end for**
7: **Return** $\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}} \leftarrow \frac{1}{n^2}\sum_{t=1}^{n^2} \widehat{\boldsymbol{\theta}}_{h,t}$

Theorem 3.2 (Privacy Guarantee of SGP_h)**.** *Algorithm 1 is (ε, δ)-differentially private.*

The privacy guarantee follows from Bassily et al. (2020, Theorem 5.1).

Now, we come to analyze the excess generalization risk

$$\mathcal{R}(\mathsf{SGP}_h; \mathbb{P}) \le \mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}})\right] + \mathbb{E}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] + \frac{1}{2} h \kappa_1.$$

It suffices to upper bound three terms separately. The key step is to show the uniform stability of Algorithm SGP_h w.r.t. ℓ. Since ℓ is L-Lipschitz continuous, it remains to control the expected distance between two returned vectors $\widehat{\boldsymbol{\theta}}$ and $\widehat{\boldsymbol{\theta}}'$ trained on a pair of neighboring datasets. The

distance can be controlled if loss function is smooth, and step size is not too large. We formally state the conclusion in the following Lemma.

Lemma 3.3. *In Algorithm* SGP_h*, suppose that* $\eta \leq 2/\beta$*, where* β *is the smoonthness parameter of* ℓ_h*. Then Algorithm* SGP_h *is* $\left(\frac{L^2\eta\cdot(1+n^2)}{n}\right)$*-uniform stable with respect to* $\ell(\cdot;\boldsymbol{x},y)$ *for any* $\boldsymbol{x}, y$*.*

The proof utilizes the non-expansiveness property of gradient update rules (Hardt et al., 2016). We note that there is a related statement (Bassily et al., 2020), but the technique we used to address nonsmoonthness here is different from theirs. Combing Lemma 2.6 and Lemma 3.3, we successfully controlled the first term in the excess generalization risk. The second term can be bounded by classic risk analysis of gradient descent with a fixed step size. Moreover, bounding the third term amounts to properly choosing bandwidth h. Combining three parts, we arrive at the first main contribution of our work.

Theorem 3.4. *Suppose that* $\boldsymbol{\theta}^* \in \mathcal{B}(B_\theta)$*. Let step size* $\eta = B_\theta/\sqrt{2L^2n^3 + (d\sigma^2 + L^2 + \kappa_1\overline{K}B_x^2/2)n^2 + 2L^2n}$*, bandwidth* $h = \eta\overline{K}B_x^2/2$ *for Algorithm* SGP_h*, then, for any distribution* $\mathbb{P}$ *over* $\mathcal{X}\times\mathcal{Y}$*, we have*

$$\mathcal{R}(\mathsf{SGP}_h;\mathbb{P}) \lesssim \max\left\{\frac{1}{\sqrt{n}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right\}.$$

The Theorem confirms asymptotic optimality of Algorithm SGP_h with convolution smoothing. More importantly, the derived excess generalization risk matches the minimax risk $\Omega\left(\max\frac{1}{\sqrt{n}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$ (Bassily et al., 2019) in terms of order, implying a rate optimality of our algorithm. The rate optimality also indicates that the additional linear term introduced by convolution smoothing is controllable and manageable, and does not harm convergence rates.

3.3. Objective Perturbation Policy

We further propose a differentially private algorithm based on objective perturbation, which is formally described below. The algorithm basically is about solving a regularized convex optimization $\min_{\boldsymbol{\theta}\in\mathbb{R}^d}\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\boldsymbol{\theta};\mathcal{D}) := \widehat{\mathcal{L}}_h(\boldsymbol{\theta};\mathcal{D}) + \lambda\|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top\boldsymbol{\theta}}{n}$ with a perturbed objective function by a noise vector $\boldsymbol{b}$. It is noteworthy that the objective perturbation policy is applied to smoothed function $\widehat{\mathcal{L}}_h$ instead of the original function $\widehat{\mathcal{L}}$. As a result, the algorithm preserves DP.

Theorem 3.5. *Algorithm 2 is* (ε,δ)*-differentially private.*

Moreover, the excess generalization risk converges to 0.

Theorem 3.6. *Assume* $\boldsymbol{\theta}^* \in \mathcal{B}(B_\theta)$*. In Algorithm* OP_h*, if we set* $\lambda = \frac{1}{B_\theta}\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2} + \frac{\kappa_1\overline{K}B_x^2}{2n\varepsilon}}$*,* $h = \frac{\overline{K}B_x^2}{\lambda n\varepsilon}$*, then, for any distribution* $\mathbb{P}$ *over* $\mathcal{X}\times\mathcal{Y}$*, its excess generalization risk satisfies*

$$\mathcal{R}(\mathsf{OP}_h;\mathbb{P}) \lesssim \max\left\{\frac{1}{\sqrt{n\varepsilon}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right\}.$$

Algorithm 2 Objective Perturbation (OP_h)

Input: Private dataset $\mathcal{D}$, privacy parameters $\varepsilon > 0, \delta \geq 0$, kernel function K with bandwidth $h > 0$, Lipschitz parameter $L = \bar{r}B_x$, smoothness parameter $\beta = \overline{K}B_x^2/h$, variance $\sigma^2 = L^2\left(8\ln(2/\delta) + 4\varepsilon\right)/\varepsilon^2$

1: Set any $\lambda \geq \frac{\beta}{n\varepsilon}$
2: Sample a noise vector $\boldsymbol{b} \sim \mathcal{N}(0, \sigma^2\mathbf{I}_{d\times d})$
3: $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b}) \leftarrow \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d}\widehat{\mathcal{L}}_h(\boldsymbol{\theta};\mathcal{D}) + \lambda\|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top\boldsymbol{\theta}}{n}$
4: **Return:** $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})$

The proof follows the same proof idea for Theorem 3.4. Specifically, uniform stability is a straightforward result because of the regularizer $\|\cdot\|_2^2$; and shrinking ERM can be derived from empirical risk analysis for regularized optimization problems. Unfortunately, the derived upper bound on excess generalization risk is not rate-optimal. An extra $\sqrt{\varepsilon}$ appears in the first term of the maximum term. Intuitively, it comes from convolution smoothing and our setting of $h \asymp 1/(\lambda n\varepsilon)$. However, one crucial advantage of objective perturbation is that it always returns a minimizer, and this advantage will play an irreplaceable role in private quantile regression.

Computational Tractability: Solving the strongly convex optimization problem in Algorithm OP_h Step 3 with gradient descent based methods are computationally efficient: gradients and Hessian matrix have closed-forms. Instead, if one is interested in solving it with an off-the-shelf solver, such as Gurobi or Mosek, there is a potential tractability issue. Since regularizer $\|\cdot\|_2$ is second-order-cone representable, and $\boldsymbol{b}^\top\boldsymbol{\theta}$ is merely a linear term, the only issue lies in representing $\widehat{\mathcal{L}}_h$ in a solver-admissible way. We note that $\widehat{\mathcal{L}}_h(\boldsymbol{\theta};\mathcal{D}) = \frac{1}{n}\sum_{i=1}^n \ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i)$ and $\ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i) = c_h(y_i - \boldsymbol{\theta}^\top\boldsymbol{x}_i)$, it remains to show the epigraph of the integral term in $c_h(\cdot)$, see Eq.(2), can be reformulated into a solver-admissible way. However, among all four kernels in Table 1, only Logistic kernel is qualified, since $\int_{-\infty}^{\infty}|u/h + v|\,K(v)\,dv = u + 2\ln(1+e^{-u})$, and its epigraph can be reformulated into exponential cones:

$$\left\{(u,t)\in\mathbb{R}^2 \middle| u + 2\ln\left(1+e^{-u}\right) \leq t\right\} = \left\{(u,t)\in\mathbb{R}^2 \middle| \begin{array}{l} \exists(v,x,y)\in\mathbb{R}^3: \\ x+y\leq 1 \\ (x,1,-u-v)\in K_{exp} \\ (y,1,-v)\in K_{exp} \\ 2v\leq t-u \end{array}\right\},$$

where $K_{exp} = \left\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 \geq x_2 e^{x_3/x_2}, x_2 > 0\right\}$ is an exponential cone. Convex optimization with exponential cones can be efficiently solved to optimum by commercial solvers, such as Mosek. Therefore, adopting Logistic kernel and implementing OP_h Step 3 with Mosek will not cause computational issues.

4. Applications

4.1. The Private Quantile Regression

Linear quantile regression is one of the most basic and important method to understand distributional information. We consider a linear data generating model taking the form $y = \langle \boldsymbol{\theta}^*, \boldsymbol{x} \rangle + \epsilon(\boldsymbol{x})$, where $\boldsymbol{\theta}^*$ is a d-dimensional coefficient vector, $\boldsymbol{x} \in \mathcal{B}(B_x)$ is the observable feature, and without loss of generality, we assume $x_1 \equiv 1$ to incorporate intercept term and assume $\mathbb{E}[x_i] = 0, \forall i = 2, \ldots, d$. The random noise $\epsilon(\boldsymbol{x})$ has a conditional Cumulative Distribution Function (CDF) $F_{y|\boldsymbol{x}}(\cdot)$ with $F^{-1}_{y|\boldsymbol{x}}(r) = 0$ for a specific quantile level $r \in (0, 1)$, which is a common assumption in quantile regression (Koenker & Hallock, 2001) (He et al., 2021).

As pointed out by (Koenker & Hallock, 2001), the quantile estimator $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}} \left[c(y - \langle \boldsymbol{\theta}, \boldsymbol{x} \rangle)\right]$, where $c(u) = ru^+ + (1-r)(-u)^+$, is optimal to minimize the generalization risk. The formulation is exactly the same as the stochastic convex optimization we considered in last section. Thus, we can employ algorithms in Section 3 to generate differentially private quantile estimators. Since Algorithm OP_h always returns a private minimizer, we restrict our attention to OP_h only.

In private quantile regression, one measurement of interest is the expected estimation error $\mathbb{E}_{\mathsf{OP}}\left[\left\|\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}_h - \boldsymbol{\theta}^*\right\|_2\right]$ between private estimator $\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}_h$ and the groundtruth $\boldsymbol{\theta}^*$. Denote $\widehat{\boldsymbol{\theta}}^{\#} = \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \widehat{\mathcal{L}}^{\#}_h(\boldsymbol{\theta}) := \widehat{\mathcal{L}}_h(\boldsymbol{\theta}) + \lambda \|\boldsymbol{\theta}\|_2^2$, and $\boldsymbol{\theta}^*_h = \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}_h(\boldsymbol{\theta})$, Then, the estimation error can be decomposed into three parts,

$$\mathbb{E}_{\mathsf{OP}}\left[\left\|\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}_h - \boldsymbol{\theta}^*\right\|_2\right] \leq \mathbb{E}_{\mathsf{OP}}\left[\left\|\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}_h - \widehat{\boldsymbol{\theta}}^{\#}_h\right\|_2\right] + \left\|\widehat{\boldsymbol{\theta}}^{\#}_h - \boldsymbol{\theta}^*_h\right\|_2 + \left\|\boldsymbol{\theta}^*_h - \boldsymbol{\theta}^*\right\|_2, \forall \mathcal{D},$$

where three parts are privacy-induced error, estimation error due to sampling, and approximation error due to convolution smoothing, respectively.

Before proceeding, we make some mild assumptions on the true data generating process.

Assumption 4.1. The unknown underlying joint distribution $\mathbb{P}$ of $(\boldsymbol{x}, \epsilon(\boldsymbol{x}))$ satisfies

- (**Lipschitz pdf**): For any $\boldsymbol{x} \in \mathcal{X}$, the conditional pdf $f_{\epsilon|\boldsymbol{x}}$ of $\epsilon(\boldsymbol{x})$ exists, and is l_1-Lipschitz continuous.
- (**Locally Strictly Positive Slope**): There exists a constant $\underline{f} > 0$ such that $f_{\epsilon|\boldsymbol{x}}(0) \geq \underline{f}$, for all $\boldsymbol{x} \in \mathcal{X}$.
- (**Widely Spread $\boldsymbol{x}$**): The matrix $\boldsymbol{\Sigma} := \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\boldsymbol{x}^\top\right] \succ \mathbf{0}$ is positive definite, and has a minimal eigenvalue $\rho_1 > 0$.

The first assumption is mild as it at least admits Gaussian noise ϵ that is independent of $\boldsymbol{x}$. The second assumption ensures uniqueness of $\boldsymbol{\theta}^*$. The last assumption is common in literature to help improve estimation accuracy as one can expect to observe $\boldsymbol{x}$ along all directions with reasonable probabilities. Now, we are ready to bound three terms.

Lemma 4.2. *Under Assumption 4.1, and let bandwidth $h > 0$ is small enough such that $\underline{f} - hl_1(\kappa_1 + \sqrt{B_x^3\kappa_2}/\rho_1) > 0$. Then, we have*

$$\|\boldsymbol{\theta}^*_h - \boldsymbol{\theta}^*\|_2 \leq \frac{h^2 l_1 \kappa_2}{\rho_1\left(\underline{f} - hl_1\kappa_1\right)}.$$

This Lemma confirms that convolution smoothing does not significantly affect optimality, and the gap reduces to 0 at a quadratic rate in h. The proof exploits the special structure of the quantile loss function.

Lemma 4.3. *For any dataset $\mathcal{D}$, if $\lambda \asymp \sqrt{\frac{1}{n\varepsilon} + \frac{d\sigma^2}{n^2}}$ where σ^2 is the variance of noise $\mathbf{b}$, then the estimator $\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}_h$ given by Algorithm OP_h satisfies*

$$\mathbb{E}_{\mathsf{OP}}\left[\left\|\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}_h - \widehat{\boldsymbol{\theta}}^{\#}_h\right\|_2\right] \lesssim \sqrt{\frac{d\ln(1/\delta)}{n\varepsilon}}.$$

The proof exploits strong convexity of $\widehat{\mathcal{L}}^{\mathsf{OP}}_h$.

Lemma 4.4. *Under Assumption 4.1, and if bandwidth $h^2 = o(1/\sqrt{n})$ and $\lambda = o(1)$, then with probability at least $1-\beta$ over random draw of dataset $\mathcal{D}$, we have*

$$\left\|\widehat{\boldsymbol{\theta}}^{\#}_h - \boldsymbol{\theta}^*_h\right\|_2 \leq O\left(\frac{l_1\kappa_2}{\rho_1\underline{f}}h^2 + \frac{L}{\rho_1\underline{f}}\sqrt{\frac{d + \ln(1/\beta)}{n}}\right).$$

The proof sketch is as follows. We first notice that proving the Lemma is equivalent to proving $\widehat{D}^{\#}_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*_h) := \widehat{\mathcal{L}}^{\#}_h(\boldsymbol{\theta}^*_h + \boldsymbol{\delta}) - \widehat{\mathcal{L}}^{\#}_h(\boldsymbol{\theta}^*_h) > 0, \forall \boldsymbol{\delta} \in \partial\mathcal{B}(r_0) := \{\boldsymbol{\delta} \in \mathbb{R}^d : \|\boldsymbol{\delta}\|_2 = r_0\}$ with a radius r_0 chosen as the value in big-O symbol. Then we lower bound $\widehat{D}^{\#}_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*_h)$ by a positive term minus some terms. And these subtrahend terms can be further upper bounded through Rademacher Complexity and covering arguments. With a vanishing bandwidth h and λ, the overall lower bound is finally found to be strictly positive with high probability. Combing three Lemmas, we are able to upper bound the estimation error.

Theorem 4.5. *Let K be a kernel function satisfying Condition 1, and assume privacy parameter $\varepsilon = \omega(1/\sqrt{n})$. In Algorithm OP_h, if $\lambda = \Theta\left(\sqrt{\frac{1}{n\varepsilon} + \frac{d\sigma^2}{n^2}}\right)$ and $h = \Theta\left(\frac{1}{\varepsilon n \lambda}\right)$ as the same values in Theorem 3.6, then for any distribution $\mathbb{P}$ that satisfies Assumption 4.1, with probability at least $1-\beta, \forall \beta \in (0,1)$ over the random draw of dataset $\mathcal{D}$, the private estimator $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$ obtained from Algorithm OP_h satisfies*

$$\mathbb{E}_{\boldsymbol{b}}\left[\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}} - \boldsymbol{\theta}^*\right\|_2\right] \leq O\left(\max\left\{\sqrt{\frac{d}{n}}, \sqrt{\frac{d\ln(1/\delta)}{n\varepsilon}}\right\}\right)$$

The Theorem gives a high-probability upper bound on estimation error w.r.t. $\|\cdot\|_2$, and the bound holds even for high-dimensional cases where $d \gg n$. The requirement on ε ensures $h^2 = o(1/\sqrt{n})$ in Lemma 4.4. Dependencies on other constants are omitted. It is well know that, by Bayes risk arguments, the statistical estimation error is at order $O(\sqrt{d/n})$. Our derived bound introduces an extra term involving privacy parameters. To our best knowledge, it is the first time to characterize the estimation error of a private quantile estimator in DP stochastic convex optimization context.

4.2. The Private Newsvendor Problem

The Newsvendor problem is the most fundamental problem in inventory control. A newsvendor must decide an order quantity q before daily demand y is realized. After demand is realized, if $q \leq y$, an underage cost $c_u \cdot (y-q)$ will occur; if $q > y$, an overage cost $c_o \cdot (q-y)$ will occure, where $c_u, c_o > 0$ are unit underage cost and unit overage cost. The newsvendor aims to minimize the generalization cost $\min_{q\in\mathbb{R}} \mathbb{E}_y\left[C(y-q)\right]$ with $C(u) := c_u \cdot u^+ + c_o \cdot (-u)^+ = (c_u + c_o)\cdot c(u)$, where $c(u)$ is the $\frac{c_u}{c_u+c_o}$-quantile loss function. The private newsvendor problem additionally requires the order quantity q to be differentially private. When demand y follows the same linear model in Section 4.1, we are safe to restrict ordering policy to a linear policy taking the form $q = \widehat{\boldsymbol{\theta}}^\top \boldsymbol{x}$. Thus, solving a private newsvendor problem amounts to finding a private estimator $\widehat{\boldsymbol{\theta}}$ and to minimize the excess generalization cost

$$\mathcal{C}(\widehat{\boldsymbol{\theta}};\mathbb{P}) := (c_u + c_o)\cdot\Big[\mathbb{E}_{\widehat{\boldsymbol{\theta}},(\boldsymbol{x},y)\sim\mathbb{P}}\left[c(y - \widehat{\boldsymbol{\theta}}^\top \boldsymbol{x})\right] - \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[c(y - \boldsymbol{\theta}^{*\top}\boldsymbol{x})\right]\Big],$$

Note that, the excess generalization cost is the same as that of a stochastic convex optimization under a quantile loss function, up to a constant factor $c_u + c_o$. Hence, proposed algorithms in Section 3 are all suitable for the private newsvendor problem.

Theorem 4.6. *1. Running Algorithm SGP_h with same settings as in Theorem 3.4 for the private newsvendor problem yields*

$$\mathcal{C}(\mathsf{SGP}_h;\mathbb{P}) \lesssim (c_u + c_o)\cdot\max\left\{\frac{1}{\sqrt{n}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right\}.$$

2. Running Algorithm OP_h with same settings as in Theorem 3.6 for the private newsvendor problem yields

$$\mathcal{C}(\mathsf{OP}_h;\mathbb{P}) \lesssim (c_u + c_o)\cdot\max\left\{\frac{1}{\sqrt{n\varepsilon}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right\}.$$

These conclusions are directly corollaries of conclusions in Section 3, and no proof needed. Moreover, the differential privacy of order quantity q is ensured by post-processing lemma of DP.

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A. Proofs for Section 3

A.1. Proof of Lemma 3.1

Proof. Gradients and Hessian matrix in part 1 of this Lemma can be easily calculated from function $c_h(\cdot)$.

As for part 2, the L-Lipschitz continuity is by noticing $\sup_{\boldsymbol{\theta},\boldsymbol{x},y} \|\nabla \ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\|_2 = \bar{r}B_x$; the β-smoothness is by noticing the supremum of maximum eigenvalue of Hessian matrix $\sup_{\boldsymbol{\theta};\boldsymbol{x},y} \lambda_{\max}\left(\nabla^2 \ell_h(\boldsymbol{\theta};\boldsymbol{x},y)\right) = \sup_{\boldsymbol{\theta};\boldsymbol{x},y} \lambda_{\max}\left(K_h\left(y-\boldsymbol{\theta}^\top \boldsymbol{x}\right)\cdot \boldsymbol{x}\boldsymbol{x}^\top\right) \le \overline{K}B_x^2/h$.

To prove part 3, it is equivalent to show $c(u) \le c_h(u) \le c(u) + \frac{1}{2}h\kappa_1, \forall u \in \mathbb{R}$. Recall that $c(u) = |u|/2 + (r-1/2)u$. Let $g_h(u) := c_h(u) - c(u)$, we have $g_h'(u) = \mathcal{K}_h(u) \ge 0, \forall u < 0$ and $g_h'(u) = \mathcal{K}_h(u) - 1 \le 0, \forall u > 0$. Since $g_h(\cdot)$ is continuous on $\mathbb{R}$, it takes maximum value at $u = 0$ with $g_h(0) = c_h(0) - c(0) = \frac{h}{2}\int_{-\infty}^{\infty} |v|\, K(v)\, dv$, which proves the right-hand-side in part 3. Moreover, this upper bound is tight by our argument. It remains to show, for any $h > 0$, $\lim_{u\to\infty} g_h(u) \ge 0$ and $\lim_{u\to-\infty} g_h(u) \ge 0$. To prove this result, we can first calculate $g_h(u)$ explicitly:

$$2g_h(u) = \int_{-\infty}^{\infty} |u+vh|\, K(v)\, dv - |u| = 2h\int_{|u|/h}^{\infty} vK(v)\, dv + u\cdot\int_{-u/h}^{u/h} K(v)\, dv - |u|\,, \quad \forall u \in \mathbb{R}.$$

We notice that, when u tends to positive or negative infinity, both limits of $g_h(u)$ exist and equal 0, which completes the proof. □

A.2. Proof of Lemma 3.3

Proof. Because of L-Lipschitz continuity of ℓ, we have

$$\mathbb{E}_{\mathsf{SGP}}\left[\ell(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}(\mathcal{D});\boldsymbol{x},y) - \ell(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}(\mathcal{D}');\boldsymbol{x},y)\right] \le L\cdot\mathbb{E}\left[\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}(\mathcal{D}) - \widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}(\mathcal{D}')\right\|_2\right], \quad \forall \boldsymbol{x}, y, \mathcal{D}\sim\mathcal{D}';$$

therefore, it suffices to control the expected deviation between returned vectors trained on two neighboring datasets. For notation brevity, the superscribe ^{SGP} and dependences on $\mathcal{D}$ are omitted throughout this proof; instead, we let $\widehat{\boldsymbol{\theta}}_h := \widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}(\mathcal{D})$ and $\widehat{\boldsymbol{\theta}}_h' := \widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}(\mathcal{D}')$. We follow the same idea in Bassily et al. (2019, Lemma 3.4) to complete our proof. We use $\widehat{\boldsymbol{\theta}}_{h,t}$ and $\widehat{\boldsymbol{\theta}}_{h,t}'$ to represent vectors in t-th iteration trained on $\mathcal{D}$ and $\mathcal{D}'$, respectively. Firstly, it can be shown that, when step size $\eta \le 2/\beta$, we have

$$\mathbb{E}_{\mathsf{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}_{h,t}'\right\|_2\right] \le \frac{2L\eta t}{n}, \quad \forall t = 1,\ldots,n^2. \tag{3}$$

This can be proved by induction. When $t = 1$, by the setting of initial points, obviously it is true. Then suppose that it is true for t-th iteration, it remains to check $(t+1)$-th iteration. Let us fix a sequence of noise vector $\{\boldsymbol{w}_t\}_{t=1}^{n^2}$, then,

$$\begin{aligned}\left\|\widehat{\boldsymbol{\theta}}_{h,t+1} - \widehat{\boldsymbol{\theta}}_{h,t+1}'\right\|_2 &= \left\|\left(\widehat{\boldsymbol{\theta}}_{h,t} - \eta(\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t};\boldsymbol{x}_{(t)},y_{(t)}) + \boldsymbol{w}_t)\right) - \left(\widehat{\boldsymbol{\theta}}_{h,t}' - \eta(\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}';\boldsymbol{x}_{(t)}',y_{(t)}') + \boldsymbol{w}_t)\right)\right\|_2 \\ &= \left\|\left(\widehat{\boldsymbol{\theta}}_{h,t} - \eta\cdot\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t};\boldsymbol{x}_{(t)},y_{(t)})\right) - \left(\widehat{\boldsymbol{\theta}}_{h,t}' - \eta\cdot\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}';\boldsymbol{x}_{(t)}',y_{(t)}')\right)\right\|_2\end{aligned} \tag{4}$$

Notice that $(\boldsymbol{x}_{(t)}, y_{(t)})$ and $(\boldsymbol{x}_{(t)}', y_{(t)}')$ are uniformly drawn from $\mathcal{D}$ and $\mathcal{D}'$, it implies that, with probability $1/n$, we have $(\boldsymbol{x}_{(t)}, y_{(t)}) \ne (\boldsymbol{x}_{(t)}', y_{(t)}')$, and it takes equality with probability $1 - 1/n$. When equal, gradient update rule is 1-expansive, i.e., (4) $\le \left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}_{h,t}'\right\|_2$ (Hardt et al. (2016, Lemma 3.6)). Consequently, taking expectation over the randomness of algorithm, we have

$$\begin{aligned}\mathbb{E}_{\mathsf{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t+1} - \widehat{\boldsymbol{\theta}}_{h,t+1}'\right\|_2\right] &\le \left(1-\frac{1}{n}\right)\mathbb{E}_{\mathsf{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}_{h,t}'\right\|_2\right] + \frac{1}{n}\mathbb{E}_{\mathsf{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}_{h,t}'\right\|_2 + 2\eta L\right] \\ &\le \mathbb{E}_{\mathsf{SGP}}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}_{h,t}'\right\|_2\right] + \frac{2\eta L}{n} \\ &\le \frac{2\eta L(t+1)}{n}.\end{aligned}$$

Therefore, by induction, our argument is true. Recall that $\widehat{\boldsymbol{\theta}}_h = \frac{1}{n^2}\sum_{t=1}^{n^2}\widehat{\boldsymbol{\theta}}_{h,t}$ is the averaged vector over all iterations, thus,

$$\mathbb{E}\left[\left\|\widehat{\boldsymbol{\theta}}_h - \widehat{\boldsymbol{\theta}}'_h\right\|_2\right] \le \frac{1}{n^2}\sum_{t=1}^{n^2}\mathbb{E}\left[\left\|\widehat{\boldsymbol{\theta}}_{h,t} - \widehat{\boldsymbol{\theta}}'_{h,t}\right\|_2\right] \le \frac{1}{n^2}\sum_{t=1}^{n^2}\frac{2\eta L t}{n} = \frac{(1+n^2)\eta L}{n}$$

□

A.3. Proof of Theorem 3.4

Proof. The excess generalization risk is

$$\mathcal{R}(\mathsf{SGP}_h;\mathbb{P}) \le \mathbb{E}_{\mathcal{D},\mathsf{SGP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}})\right] + \mathbb{E}_{\mathcal{D},\mathsf{SGP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] + \frac{1}{2}h\kappa_1.$$

By Lemma 3.3, we know that $\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}$ is uniformly stable w.r.t. ℓ, and therefore according to Lemma 2.6, the first term is upper bounded by $\frac{L^2\eta\cdot(1+n^2)}{n}$. Now, we fix a dataset $\mathcal{D}$ and come to bound the second term:

$$\begin{aligned}
\mathbb{E}_{\mathsf{SGP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] &\le \frac{1}{n^2}\cdot\mathbb{E}_{\mathsf{SGP}}\left[\sum_{t=1}^{n^2}\left(\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right)\right] \\
&\le \frac{1}{n^2}\cdot\mathbb{E}_{\mathsf{SGP}}\left[\sum_{t=1}^{n^2}\left\langle\widehat{\boldsymbol{\theta}}_{h,t} - \boldsymbol{\theta}^*, \nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t})\right\rangle\right]. \\
&= \frac{1}{n^2}\cdot\mathbb{E}_{\mathsf{SGP}}\left[\sum_{t=1}^{n^2}\left\langle\widehat{\boldsymbol{\theta}}_{h,t} - \boldsymbol{\theta}^*, \nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}, \boldsymbol{x}_{(t)}, y_{(t)}) + \boldsymbol{w}_t\right\rangle\right] \\
&\le \frac{1}{n^2}\cdot\mathbb{E}_{\mathsf{SGP}}\left[\frac{\|\boldsymbol{\theta}^*\|_2^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{n^2}\left\|\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t}, \boldsymbol{x}_{(t)}, y_{(t)}) + \boldsymbol{w}_t\right\|_2^2\right], \\
&\le \frac{\|\boldsymbol{\theta}^*\|_2^2}{2n^2\eta} + \frac{\eta L^2}{2} + \frac{\eta d\sigma^2}{2},
\end{aligned}$$

where first two lines are due to convexity of $\widehat{\mathcal{L}}_h$; the third line comes from the fact that $\boldsymbol{w}_t$ is drawn from a zero-mean Gaussian distribution and is independent of $\widehat{\boldsymbol{\theta}}_{h,t}$, and $\nabla\ell_h(\widehat{\boldsymbol{\theta}}_{h,t};\boldsymbol{x}_{(t)}, y_{(t)})$ is an unbiased gradient to $\nabla\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t})$; the forth line follows from a classic gradient descent analysis (Shalev-Shwartz & Ben-David (2014, Lemma 14.1)) and from the gradient descent update rule in our algorithm; the last line is due to L-Lipschitz continuity of ℓ_h and Gaussian vector's upper bounds.

Plugging back into the risk expression, letting $h = \eta\overline{K}B_x^2/2$ (which ensures $\eta\beta = 2$, making Lemma 3.3 hold), and replacing $\|\boldsymbol{\theta}^*\|_2$ with B_θ, we obtain

$$\mathcal{R}(\mathsf{SGP}_h;\mathbb{P}) \le \frac{L^2\eta\cdot(1+n^2)}{n} + \frac{B_\theta^2}{2n^2\eta} + \frac{\eta L^2}{2} + \frac{\eta d\sigma^2}{2} + \frac{\eta\kappa_1\overline{K}B_x^2}{4}, \quad \forall\eta \ge 0.$$

It is easy to find $\eta^* = B_\theta/\sqrt{2L^2n^3 + (d\sigma^2 + L^2 + \kappa_1\overline{K}B_x^2/2)n^2 + 2L^2n}$ minimizes the r.h.s, and gives

$$\mathcal{R}(\mathsf{SGP}_h;\mathbb{P}) \le B_\theta L\sqrt{\frac{2}{n} + \frac{1 + 8d\ln(1/\delta)/\varepsilon^2 + \kappa_1\overline{K}B_x^2/2}{n^2} + \frac{2}{n^3}} \lesssim B_\theta L\cdot\max\left\{\frac{1}{\sqrt{n}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right\}.$$

Lastly, we note that the above reasoning can be applied to any distribution $\mathbb{P}$ over $\mathcal{X}\times\mathcal{Y}$, which completes the proof. □

A.4. Proof of Theorem 3.5

Proof. The Theorem directly follows from Kifer et al. (2012, Theorem 2). According to Kifer et al. (2012, Theorem 2), the minimizer $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})$ will satisfy (ε,δ)-DP if[1] (1) Hessian matrix $\nabla^2\widehat{\mathcal{L}}_h$ is continuous and at most rank-1 (2) loss function ℓ_h

[1] conditions are rephrased to be consistent with our context

is L-Liptschiz continuous and β-smooth (3) noise vector $\boldsymbol{b}$ is sampled from a multivariate Gaussian $\mathcal{N}(0, \sigma^2 \mathbf{I}_{d\times d})$ with variance $\sigma^2 \geq L^2 \left(8 \ln\left(2/\delta\right) + 4\varepsilon\right)/\varepsilon^2$; (4) and the regularization coefficient $\lambda \geq \beta/(n\varepsilon)$. It is easy to check that all four conditions are satisfied in our settings. □

A.5. Proof of Theorem 3.6

Before proceeding, we summarize all notations in Table 2 for analyzing objective perturbation policy.

Table 2. Notations for Objective Perturbation policy

Type	Description	Abbr.	Functions	Minimizers
Empirical	ERM	$\widehat{\mathcal{L}}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$	$\widehat{\boldsymbol{\theta}}$
	Regularized ERM	$\widehat{\mathcal{L}}^{\#}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i) + \lambda \left\|\boldsymbol{\theta}\right\|_2^2$	$\widehat{\boldsymbol{\theta}}^{\#}$
	Private regularized ERM	$\widehat{\mathcal{L}}^{\mathsf{OP}}(\boldsymbol{\theta}; \boldsymbol{b})$	$\frac{1}{n}\sum_{i=1}^n \ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i) + \lambda \left\|\boldsymbol{\theta}\right\|_2^2 + \frac{\boldsymbol{b}^\top \boldsymbol{\theta}}{n}$	$\widehat{\boldsymbol{\theta}}^{\mathsf{OP}}$
Smoothed	ERM	$\widehat{\mathcal{L}}_h(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^n \ell_h(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)$	$\widehat{\boldsymbol{\theta}}_h$
	Regularized ERM	$\widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta})$	$\frac{1}{n}\sum_{i=1}^n \ell_h(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i) + \lambda \left\|\boldsymbol{\theta}\right\|_2^2$	$\widehat{\boldsymbol{\theta}}_h^{\#}$
	Private regularized ERM	$\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\boldsymbol{\theta}; \boldsymbol{b})$	$\frac{1}{n}\sum_{i=1}^n \ell_h(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i) + \lambda \left\|\boldsymbol{\theta}\right\|_2^2 + \frac{\boldsymbol{b}^\top \boldsymbol{\theta}}{n}$	$\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$

Proof. We separately bound excess generalization risk's three terms:

$$\mathcal{R}(\mathsf{OP}_h; \mathbb{P}) \leq \mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right] + \mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] + \frac{1}{2}h\kappa_1,$$

where $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$ depends on both dataset $\mathcal{D}$ and noise vector $\boldsymbol{b}$. We notationally suppress the dependences unless explicitly manipulating $\mathcal{D}$ and $\boldsymbol{b}$.

Because L-Lipschitz continuity of ℓ implies $\ell(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D})) - \ell(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')) \leq L\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}) - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')\right\|_2$, it suffices to control the distance between minimizers trained on two neighboring datasets $\mathcal{D} \sim \mathcal{D}'$. By classic stability analysis for strongly convex optimization problem with a regularizer $\|\cdot\|_2^2$ (Bousquet & Elisseeff, 2002), we have $\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}) - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\mathcal{D}')\right\|_2 \leq \frac{2L}{n\lambda}$, $\forall \mathcal{D} \sim \mathcal{D}', \boldsymbol{b}$. Therefore, Algorithm OP_h is $\frac{2L^2}{n\lambda}$-uniform stable w.r.t. ℓ. By Lemma 2.6, we know that $\mathbb{E}_{\mathcal{D},\mathsf{OP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \widehat{\mathcal{L}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}})\right] \leq \frac{2L^2}{n\lambda}$.

We now fix a dataset $\mathcal{D}$ and noise vector $\boldsymbol{b}$, and come to bound $\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)$. By strong convexity of $\widehat{\mathcal{L}}_h^{\mathsf{OP}}$ and Cauchy's inequality, we have

$$\lambda\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2^2 \leq \widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\#}) - \widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) = \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\#}) - \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) + \frac{\boldsymbol{b}^\top \widehat{\boldsymbol{\theta}}_h^{\#}}{n} - \frac{\boldsymbol{b}^\top \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}}{n} \leq \frac{\|\boldsymbol{b}\|_2 \left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2}{n},$$

which gives $\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2 \leq \frac{\|\boldsymbol{b}\|_2}{n\lambda}$. We further notice that

$$\begin{aligned}
\widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*) &\leq \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\#}) + \lambda\left(\|\boldsymbol{\theta}^*\|_2^2 - \left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2^2\right) \\
&\leq \left[\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \frac{\boldsymbol{b}^\top \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}}{n}\right] - \left[\widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}) - \frac{\boldsymbol{b}^\top \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}}{n}\right] + \lambda\|\boldsymbol{\theta}^*\|_2^2 \\
&\leq \frac{\|\boldsymbol{b}\|_2 \cdot \left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})\right\|_2}{n} + \lambda\|\boldsymbol{\theta}^*\|_2^2 \\
&\leq \frac{\|\boldsymbol{b}\|_2^2}{n^2\lambda} + \lambda\|\boldsymbol{\theta}^*\|_2^2,
\end{aligned}$$

where the first inequality is because $\widehat{\boldsymbol{\theta}}_h^{\#}$ is the minimizer to $\widehat{\mathcal{L}}_h^{\#}$; the second inequality is by plugging in minimizer $\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}$ into the second bracket, and dropping a negative term $-\lambda \left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}\right\|_2^2$. The reasoning here holds for any $\mathcal{D}$ and $\boldsymbol{b}$.

Substituting both upper bounds back to $\mathcal{R}(\mathsf{OP}_h;\mathbb{P})$, we obtain

$$\mathcal{R}(\mathsf{OP}_h;\mathbb{P}) \leq \frac{2L^2}{n\lambda} + \frac{\mathbb{E}_{\boldsymbol{b}}\left[\|\boldsymbol{b}\|_2^2\right]}{n^2\lambda} + \lambda\|\boldsymbol{\theta}^*\|_2^2 + \frac{1}{2}h\kappa_1, \quad \forall \lambda > 0, h > 0.$$

Lastly, replacing $\|\boldsymbol{\theta}^*\|_2$ with B_θ, setting $h = \frac{\overline{K}B_x^2}{\lambda n\varepsilon}$, and optimizing over λ, we get the optimal $\lambda^* = \frac{1}{B_\theta}\sqrt{\frac{2L^2}{n} + \frac{d\sigma^2}{n^2} + \frac{\kappa_1\overline{K}B_x^2}{2n\varepsilon}}$, and

$$\mathcal{R}(\mathsf{OP}_h;\mathbb{P}) \leq 2B_\theta\sqrt{\frac{2L^2}{n} + \frac{dL^2(8\ln(2/\delta)+4\varepsilon)}{n^2\varepsilon^2} + \frac{\kappa_1\overline{K}B_x^2}{2n\varepsilon}} \lesssim \max\left\{\frac{1}{\sqrt{n\varepsilon}}, \frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right\}.$$

□

B. Proofs for Section 4

B.1. Proof of Lemma 4.2

Proof. We follow the same proof idea for Proposition 4.1 in (He et al., 2021), but our assumptions and desired statement are different from theirs. We first compute the following expected value conditional on $\boldsymbol{x}$,

$$\begin{aligned}
\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}(-\epsilon/h)\middle|\boldsymbol{x}\right] &= \int_{-\infty}^{\infty} \mathcal{K}(-t/h)\, dF_{\epsilon|\boldsymbol{x}}(t) \\
&= \frac{1}{h}\int_{-\infty}^{\infty} F_{\epsilon|\boldsymbol{x}}(t)K(-t/h)\, dt && \text{(integration by parts)} \\
&= \int_{-\infty}^{\infty} F_{\epsilon|\boldsymbol{x}}(-uh)K(u)\, du && \text{(let } u = -t/h\text{)} \\
&= r + \int_{-\infty}^{\infty}\left[F_{\epsilon|\boldsymbol{x}}(-uh) - F_{\epsilon|\boldsymbol{x}}(0)\right]K(u)\, du && \left(F_{\epsilon|\boldsymbol{x}}(0) = r \text{ and } \int K = 1\right) \\
&= r + \int_{-\infty}^{\infty} K(u)\left[\int_0^{-uh} f_{\epsilon|\boldsymbol{x}}(t) - f_{\epsilon|\boldsymbol{x}}(0)\, dt\right] du \\
&\leq r + \int_{-\infty}^{\infty} K(u)\int_0^{|-uh|} l_1 t\, dt\, du && (f_{\epsilon|\boldsymbol{x}} \text{ is } l_1\text{-Lipschitz}) \\
&= r + \frac{1}{2}h^2 l_1 \underbrace{\int_{-\infty}^{\infty} u^2 K(u)\, du}_{=:\kappa_2}. && (5)
\end{aligned}$$

By above calculation, we can upper bound $\nabla\mathcal{L}_h(\boldsymbol{\theta}^*)$ as below:

$$\begin{aligned}
\nabla\mathcal{L}_h(\boldsymbol{\theta}^*) &= \mathbb{E}_{y,\boldsymbol{x}}\left[\left[\mathcal{K}_h({\boldsymbol{\theta}^*}^\top\boldsymbol{x} - y) - r\right]\boldsymbol{x}\right] \\
&= \mathbb{E}_{\boldsymbol{x}}\left[\mathbb{E}_{\epsilon|\boldsymbol{x}}\left[\mathcal{K}(-\epsilon/h) - r\middle|\boldsymbol{x}\right]\cdot\boldsymbol{x}\right] \\
&\leq \frac{1}{2}h^2 l_1\kappa_2\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{x}]. && \text{(by (5))} && (6)
\end{aligned}$$

Since we assume $\mathbb{E}[\boldsymbol{x}] = [1, 0, \ldots, 0]$, naturally we have

$$\|\nabla\mathcal{L}_h(\boldsymbol{\theta}^*)\|_2 \leq \frac{1}{2}h^2 l_1\kappa_2. \tag{7}$$

Thus, we can upper bound the following value

$$\begin{aligned}\langle \nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) - \nabla \mathcal{L}_h(\boldsymbol{\theta}^*), \boldsymbol{\theta}_h^* - \boldsymbol{\theta}^* \rangle &\leq \|\nabla \mathcal{L}_h(\boldsymbol{\theta}^*)\|_2 \cdot \|\boldsymbol{\theta}_h^* - \boldsymbol{\theta}^*\|_2 && (\text{since } \nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) = \mathbf{0}) \\ &\leq \frac{1}{2} h^2 l_1 \kappa_2 \cdot \|\boldsymbol{\theta}_h^* - \boldsymbol{\theta}^*\|_2 . \end{aligned} \tag{8}$$

In addition, we can lower bound the left-hand-side of (8), but before showing that, we first lower bound the Hessian matrix $\nabla^2 \mathcal{L}_h(\boldsymbol{\theta}), \forall \boldsymbol{\theta}$.

It can be shown that, for a given $\boldsymbol{\theta}$,

$$\begin{aligned}\mathbb{E}_{y|\boldsymbol{x}} \left[K_h(y - \boldsymbol{\theta}^\top \boldsymbol{x}) \middle| \boldsymbol{x} \right] &= \int_{-\infty}^{\infty} \frac{1}{h} K\left(\frac{\boldsymbol{\theta}^{*\top} \boldsymbol{x} - \boldsymbol{\theta}^\top \boldsymbol{x} - t}{h} \right) dF_{\epsilon|\boldsymbol{x}}(t) \\ &= \int_{-\infty}^{\infty} K(u) f_{\epsilon|\boldsymbol{x}}(-\boldsymbol{\delta}^\top \boldsymbol{x} - uh)\, du && (\boldsymbol{\delta} := \boldsymbol{\theta} - \boldsymbol{\theta}^*; \text{ let } u = \frac{-\boldsymbol{\delta}^\top \boldsymbol{x} - t}{h}) \\ &\geq \int_{-\infty}^{\infty} K(u) \left(f_{\epsilon|\boldsymbol{x}}(0) - l_1 \left| \boldsymbol{\delta}^\top \boldsymbol{x} + uh \right| \right) du && (f_{\epsilon|\boldsymbol{x}} \text{ is } l_1\text{-Lipschitz}) \\ &\geq \underline{f} - h l_1 \underbrace{\int_{-\infty}^{\infty} |u|\, K(u)\, du}_{=:\kappa_1} - l_1 \left| \boldsymbol{\delta}^\top \boldsymbol{x} \right| . \end{aligned}$$

Therefore, the Hessian matrix of function $\mathcal{L}_h$ at point $\boldsymbol{\theta} \in \mathbb{R}^d$ satisfies

$$\nabla^2 \mathcal{L}_h(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x}} \left[\mathbb{E}_{y|\boldsymbol{x}} \left[K_h(y - \boldsymbol{\theta}^\top \boldsymbol{x}) \middle| \boldsymbol{x} \right] \cdot \boldsymbol{x}\boldsymbol{x}^\top \right] \succcurlyeq \mathbb{E}_{\boldsymbol{x}} \left[\left(\underline{f} - h l_1 \kappa_1 - l_1 \left| \boldsymbol{\delta}^\top \boldsymbol{x} \right| \right) \cdot \boldsymbol{x}\boldsymbol{x}^\top \right],$$

where $\boldsymbol{\delta} := \boldsymbol{\theta} - \boldsymbol{\theta}^*$ is a vector that originates from $\boldsymbol{\theta}^*$ to $\boldsymbol{\theta}$. By mean value theorem for vector-valued functions, we know that ,

$$\nabla \mathcal{L}_h(\boldsymbol{\theta}) - \nabla \mathcal{L}_h(\boldsymbol{\theta}^*) = \int_0^1 \nabla^2 \mathcal{L}_h(\boldsymbol{\theta}^* + \lambda \boldsymbol{\delta})\, d\lambda \cdot \boldsymbol{\delta}, \quad \forall \boldsymbol{\theta}.$$

By plugging the expression of Hessian matrix $\nabla^2 \mathcal{L}_h$, we are able to show that, for any finite $\boldsymbol{\delta} := \boldsymbol{\theta} - \boldsymbol{\theta}^*$,

$$\begin{aligned}\langle \nabla \mathcal{L}_h(\boldsymbol{\theta}) - \nabla \mathcal{L}_h(\boldsymbol{\theta}^*), \boldsymbol{\theta} - \boldsymbol{\theta}^* \rangle &= \boldsymbol{\delta}^\top \cdot \int_0^1 \nabla^2 \mathcal{L}_h(\boldsymbol{\theta}^* + \lambda \boldsymbol{\delta})\, d\lambda \cdot \boldsymbol{\delta} \\ &\geq \int_0^1 \mathbb{E}_{\boldsymbol{x}} \left[\left(\underline{f} - h l_1 \kappa_1 - l_1 \left| \lambda \boldsymbol{\delta}^\top \boldsymbol{x} \right| \right) \cdot \left(\boldsymbol{\delta}^\top \boldsymbol{x} \right)^2 \right] d\lambda \\ &= \mathbb{E}_{\boldsymbol{x}} \left[\int_0^1 \left(\underline{f} - h l_1 \kappa_1 - l_1 \left| \lambda \boldsymbol{\delta}^\top \boldsymbol{x} \right| \right) d\lambda \cdot \left(\boldsymbol{\delta}^\top \boldsymbol{x} \right)^2 \right] \\ &\geq \rho_1 (\underline{f} - h l_1 \kappa_1) \|\boldsymbol{\delta}\|_2^2 - \frac{1}{2} l_1 B_x^3 \|\boldsymbol{\delta}\|_2^3 , \end{aligned} \tag{9}$$

where the switch of integrals in third line holds from Fubini's Theorem. Thus, if we consider $\boldsymbol{\delta}_h^* := \boldsymbol{\theta}_h^* - \boldsymbol{\theta}^*$, we get

$$\nabla \mathcal{L}_h(\boldsymbol{\theta}_h^*) - \nabla \mathcal{L}_h(\boldsymbol{\theta}^*) \geq \rho_1 (\underline{f} - h l_1 \kappa_1) \|\boldsymbol{\delta}_h^*\|_2^2 - \frac{1}{2} l_1 B_x^3 \|\boldsymbol{\delta}_h^*\|_2^3 . \tag{10}$$

Combining upper bound (8) and lower bound (10) gives

$$\underbrace{\frac{1}{2} l_1 B_x^3}_{=:a>0} \|\boldsymbol{\delta}_h^*\|_2^3 - \underbrace{\rho_1 (\underline{f} - h l_1 \kappa_1)}_{=:b>0} \|\boldsymbol{\delta}_h^*\|_2^2 + \underbrace{\frac{1}{2} h^2 l_1 \kappa_2}_{=:c>0} \|\boldsymbol{\delta}_h^*\|_2 \geq 0,$$

solving which results in three solutions,

$$\|\boldsymbol{\delta}_h^*\|_2 \geq 0; \quad \text{or } \|\boldsymbol{\delta}_h^*\|_2 \leq \frac{2c}{b + \Delta^{1/2}}; \quad \text{or } \|\boldsymbol{\delta}_h^*\|_2 \geq \frac{b + \Delta^{1/2}}{2a},$$

provided that $\Delta := b^2 - 4ac > 0$. We can rule out the third solution by contradiction. Suppose that the third solution is true, then $\|\boldsymbol{\delta}_h^*\|_2 > b/(2a) > \sqrt{c/a}$. Since $\|\boldsymbol{\delta}_h^*\|_2 > \sqrt{c/a}$, there exists $\alpha \in (0,1)$ such that $\alpha \cdot \|\boldsymbol{\delta}_h^*\|_2 = \sqrt{c/a}$. With this α, let us denote $\boldsymbol{\theta}_\alpha := (1-\alpha)\boldsymbol{\theta}^* + \alpha\boldsymbol{\theta}_h^*$. We notice that, by (8),

$$\langle -\nabla\mathcal{L}_h(\boldsymbol{\theta}^*), \boldsymbol{\theta}_\alpha - \boldsymbol{\theta}^*\rangle = \langle \nabla\mathcal{L}_h(\boldsymbol{\theta}^*), \alpha\boldsymbol{\delta}_h^*\rangle \le c\alpha \|\boldsymbol{\delta}_h^*\|_2 . \tag{11}$$

Moreover, since $\langle \nabla\mathcal{L}_h(\boldsymbol{\theta}_\alpha), \boldsymbol{\theta}_\alpha - \boldsymbol{\theta}^*\rangle \le 0$, we also have

$$\begin{aligned}\langle -\nabla\mathcal{L}_h(\boldsymbol{\theta}^*), \boldsymbol{\theta}_\alpha - \boldsymbol{\theta}^*\rangle &\ge \langle \nabla\mathcal{L}_h(\boldsymbol{\theta}_\alpha) - \nabla\mathcal{L}_h(\boldsymbol{\theta}^*), \boldsymbol{\theta}_\alpha - \boldsymbol{\theta}^*\rangle \\ &\ge b\|\boldsymbol{\theta}_\alpha - \boldsymbol{\theta}^*\|_2^2 - a\|\boldsymbol{\theta}_\alpha - \boldsymbol{\theta}^*\|_2^3 \qquad \text{(by (9))} \\ &= \alpha^2\|\boldsymbol{\delta}_h^*\|_2^2\left(b - a\cdot\sqrt{c/a}\right) \end{aligned} \tag{12}$$

Combining (11) and (12) together and cancelling out $\alpha\|\boldsymbol{\delta}_h^*\|_2$ on both sides, we obtain

$$\alpha\|\boldsymbol{\delta}_h^*\|_2 \le \frac{c}{b-\sqrt{ac}}, \tag{13}$$

which is true without the need of changing inequality symbol as $\Delta > 0$ implying both sides are positive. However, we further have

$$\sqrt{c/a} = \alpha\|\boldsymbol{\delta}_h^*\|_2 \le \frac{c}{b-\sqrt{ac}} < \frac{c}{\sqrt{ac}} = \sqrt{c/a},$$

a contradiction. Therefore, we are safe to rule out the third solution. As a result,

$$0 \le \|\boldsymbol{\delta}_h^*\|_2 \le \frac{2c}{b+\Delta^{1/2}} \le \frac{2c}{b},$$

provided that $\Delta > 0$ and $b > 0$, for which a sufficient condition is $\underline{f} - hl_1(\kappa_1 + \sqrt{B_x^3\kappa_2}/\rho_1) > 0$. □

B.2. Proof of Lemma 4.3

Proof. By strong convexity of $\widehat{\mathcal{L}}_h^{\mathsf{OP}}$, for any fixed $\mathcal{D}$ and $\boldsymbol{b}$, we have

$$\begin{aligned}\lambda\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})\right\|_2^2 &\le \widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\#}) - \widehat{\mathcal{L}}_h^{\mathsf{OP}}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})) \\ &= \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\#}) - \widehat{\mathcal{L}}_h^{\#}(\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})) + \frac{\boldsymbol{b}^\top\widehat{\boldsymbol{\theta}}_h^{\#}}{n} - \frac{\boldsymbol{b}^\top\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})}{n} \\ &\le \frac{\|\boldsymbol{b}\|_2\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})\right\|_2}{n},\end{aligned}$$

which implies $\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})\right\|_2 \le \|\boldsymbol{b}\|_2/(n\lambda)$. And further replacing $\lambda \asymp \sqrt{\frac{1}{n\varepsilon} + \frac{d\sigma^2}{n^2}}$ and taking expectation over $\boldsymbol{b}$, we have

$$\mathbb{E}_{\boldsymbol{b}}\left[\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b}) - \widehat{\boldsymbol{\theta}}_h^{\#}\right\|_2\right] \lesssim \frac{\mathbb{E}_{\boldsymbol{b}}\left[\|\boldsymbol{b}\|_2\right]}{n\cdot\sqrt{\frac{1}{n\varepsilon} + \frac{d\sigma^2}{n^2}}} \lesssim \sqrt{\frac{d\ln(1/\delta)}{n\varepsilon}}.$$

□

B.3. Proof of Lemma 4.4

Proof. For any $\boldsymbol{\delta} \in \mathbb{R}^d$ and $\boldsymbol{\theta} \in \mathbb{R}^d$, we define the population-level zero-order Taylor expansion remainder of $\mathcal{L}_h(\boldsymbol{\theta}+\boldsymbol{\delta})$ at point $\boldsymbol{\theta}$ as $D_h(\boldsymbol{\delta},\boldsymbol{\theta}) = \mathcal{L}_h(\boldsymbol{\theta}+\boldsymbol{\delta}) - \mathcal{L}_h(\boldsymbol{\theta})$, the first-order remainder as $R_h(\boldsymbol{\delta},\boldsymbol{\theta}) = D_h(\boldsymbol{\delta},\boldsymbol{\theta}) - \nabla\mathcal{L}_h(\boldsymbol{\theta})^\top\boldsymbol{\delta}$. Correspondingly, the sample-level remainders are defined as $\widehat{D}_h(\boldsymbol{\delta},\boldsymbol{\theta}) = \widehat{\mathcal{L}}_h(\boldsymbol{\theta}+\boldsymbol{\delta}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta})$ and $\widehat{R}_h(\boldsymbol{\delta},\boldsymbol{\theta}) = \widehat{D}_h(\boldsymbol{\delta},\boldsymbol{\theta}) - \nabla\widehat{\mathcal{L}}_h(\boldsymbol{\theta})^\top\boldsymbol{\delta}$. And similarly for regularized variants, $\widehat{D}_h^{\#}(\boldsymbol{\delta},\boldsymbol{\theta}) = \widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta}+\boldsymbol{\delta}) - \widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta})$ and $\widehat{R}_h^{\#}(\boldsymbol{\delta},\boldsymbol{\theta}) = \widehat{D}_h^{\#}(\boldsymbol{\delta},\boldsymbol{\theta}) - \nabla\widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta})^\top\boldsymbol{\delta}$.

With these notations, for any $\boldsymbol{\delta} \in \mathbb{R}^d$, we have

$$\begin{aligned}\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) &= \underbrace{R_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) + \nabla \mathcal{L}_h(\boldsymbol{\theta}^*)^\top \boldsymbol{\delta} - D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*)}_{=0, \text{ by definition}} + \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) \\ &= R_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) + \nabla \mathcal{L}_h(\boldsymbol{\theta}^*)^\top \boldsymbol{\delta} - \left(D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right) - \left(\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*)\right)\end{aligned}$$

To find a qualified upper bound r_0 on $\left\|\widehat{\boldsymbol{\theta}}_h^{\#} - \boldsymbol{\theta}_h^*\right\|_2$, by Lemma 9.21 in (Wainwright, 2019), it suffices to show that a candidate r_0 satisfies $\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) > 0, \forall \boldsymbol{\delta} \in \partial\mathcal{B}(r_0) := \left\{\boldsymbol{\delta} \in \mathbb{R}^d : \|\boldsymbol{\delta}\|_2 = r_0\right\}$. Our proof strategy is to first lower bound $\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*)$ and show this lower bound is strictly greater than 0 for any $\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)$ with our choice of r_0. Note that

$$\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) \geq R_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \|\nabla \mathcal{L}_h(\boldsymbol{\theta}^*)\|_2 \|\boldsymbol{\delta}\|_2 - \left(D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right) - \left(\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*)\right). \tag{14}$$

Hence, it suffices to lower bound first term, and upper bound last three terms. Keep $\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)$ in mind, we address them one by one. For notational brevity, we use same notations as in the proof for Lemma 4.2 and denote

$$a := \frac{1}{2} l_1 B_x^3; \quad b := \rho_1(\underline{f} - h l_1 \kappa_1); \quad \text{and } c := \frac{1}{2} h^2 l_1 \kappa_2.$$

Note that all three values are positive under assumption that $\underline{f} - h l_1(\kappa_1 + \sqrt{B_x^3 \kappa_2}/\rho_1) > 0$, and $a \to \frac{1}{2} l_1 B_x^3$, $b \to \rho_1 \underline{f}$, $c \to 0$ as $h \to 0_+$.

1. Following the same argument for (9), we can show that

$$R_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) \geq \rho_1(\underline{f} - h l_1 \kappa_1) \|\boldsymbol{\delta}\|_2^2 - \frac{1}{2} l_1 B_x^3 \|\boldsymbol{\delta}\|_2^3 = -a r_0^3 + b r_0^2 \tag{15}$$

2. By (7), we know that $\|\nabla \mathcal{L}_h(\boldsymbol{\theta}^*)\|_2 \leq \frac{1}{2} h^2 l_1 \kappa_2$; thus,

$$\|\nabla \mathcal{L}_h(\boldsymbol{\theta}^*)\|_2 \|\boldsymbol{\delta}\|_2 \leq c r_0. \tag{16}$$

3. The third term $D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*)$ is well-studied in Lemma B.1. Specifically, we know from Lemma B.1 that, for any given $r_0 > 0$ and $\beta \in (0, 1)$, with high probability at least $1 - \beta$,

$$\sup_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)} \left\{D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right\} \leq 3 r_0 L \sqrt{\frac{4d + \ln(1/\beta)}{n}} + \lambda r_0^2. \tag{17}$$

4. By L-Lipschitz continuity of $\widehat{\mathcal{L}}_h$, we can show that, for any $\boldsymbol{\delta} \in \mathbb{R}^d$,

$$\begin{aligned}\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) &= \left[\widehat{\mathcal{L}}_h(\boldsymbol{\theta}^* + \boldsymbol{\delta}) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}_h^* + \boldsymbol{\delta})\right] + \left[\widehat{\mathcal{L}}_h(\boldsymbol{\theta}_h^*) - \widehat{\mathcal{L}}_h(\boldsymbol{\theta}^*)\right] \\ &\quad + \lambda \|\boldsymbol{\theta}^* + \boldsymbol{\delta}\|_2^2 - \lambda \|\boldsymbol{\theta}_h^* + \boldsymbol{\delta}\|_2^2 + \lambda \|\boldsymbol{\theta}_h^*\|_2^2 - \lambda \|\boldsymbol{\theta}^*\|_2^2 \\ &\leq 2L \cdot \|\boldsymbol{\theta}_h^* - \boldsymbol{\theta}^*\|_2 + 2\lambda (\boldsymbol{\theta}^* - \boldsymbol{\theta}_h^*)^\top \boldsymbol{\delta} \\ &\leq (2L + 2\lambda r_0) \|\boldsymbol{\theta}^* - \boldsymbol{\theta}_h^*\|_2\end{aligned} \tag{18}$$

Plugging (15), (16), (17), and (18) into (14), we obtain by rearranging that, for any $\beta \in (0, 1)$, with probability at least $1 - \beta$, the inequality holds for any $\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)$,

$$\begin{aligned}\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) &\geq -a r_0^3 + (b + \lambda) r_0^2 - \left(c + 3L\sqrt{\frac{4d + \ln(1/\beta)}{n}} + 2\lambda \|\boldsymbol{\theta}^* - \boldsymbol{\theta}_h^*\|_2\right) r_0 - 2L \|\boldsymbol{\theta}^* - \boldsymbol{\theta}_h^*\|_2 \\ &\geq -a r_0^3 + \underbrace{(b + \lambda)}_{=: u_n} r_0^2 - \left(\underbrace{(1 + 4\lambda/b)c + 3L\sqrt{\frac{4d + \ln(1/\beta)}{n}}}_{=: v_n}\right) r_0 - \underbrace{4cL/b}_{=: t_n} \quad =: g_n(r_0).\end{aligned} \tag{19}$$

The last line is a cubic function in r_0, and we use the subscript $_n$ to emphasize the dependence on n. We argue that there exists a threshold N such that when $n \geq N$, we can choose radius $r_0 = C_0 v_n/u_n$ for some constant $C_0 > 0$ so that $g_n(r_0) > 0$. To prove this argument, it is helpful to check the value $g_n(Cv_n/u_n)$ for any C,

$$\phi_n(C) := g_n(Cv_n/u_n) = -a\frac{v_n^3}{u_n^3}C^3 + \frac{v_n^2}{u_n}C^2 - \frac{v_n^2}{u_n}C - t_n,$$

another cubic function in C that decreases when $C \leq 0$, and intersects with vertical axis at negative line. Since we assume $h^2 = o(1/\sqrt{n})$ and $\lambda = o(1)$, we have

$$u_n = \Theta\left(\rho_1 \underline{f}\right), \quad v_n = \Theta\left(l_1\kappa_2 h^2 + L\sqrt{\frac{d + \ln(1/\beta)}{n}}\right), \quad \text{and } t_n = \Theta\left(\frac{l_1\kappa_2 h^2}{\rho_1 \underline{f}}\right).$$

Then, it can be shown, by relationships between roots and coefficients of cubic functions, that for $n > N$, the function ϕ_n has three roots $x_1(n) \leq 0 \leq x_2(n) < x_3(n)$, and $\phi_n(C) \geq 0, \forall C \in [x_2(n), x_3(n)]$. Moreover, $x_1(n) \to 0_-$, $x_2(n) \to 0_+$ and $x_3(n) \to \infty$, which implies that any $C_0 \in [x_2(n), x_3(n)]$ that makes ϕ_n positive will ensure all subsequent functions $\{\phi_{n+i}\}_{i=1}^{\infty}$ positive at $C = C_0$. Consequently, such C_0 makes $g_n(r_0)$ strictly positive for every larger n. Equivalently, that means $\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}_h^*) > 0, \forall \boldsymbol{\delta} \in \partial\mathcal{B}(r_0)$ with

$$r_0 = C_0 v_n/u_n = O\left(\frac{l_1\kappa_1}{\rho_1 \underline{f}}h^2 + \frac{L}{\rho_1 \underline{f}}\sqrt{\frac{d + \ln(1/\beta)}{n}}\right),$$

for large enough n. It is noteworthy that λ does not influence the radius since its term is dominated by c in v_n. Paraphrasing above statement with big-O statement and plugging b, c, λ back into v_n and u_n, we obtain the desired Lemma. □

Lemma B.1. *Given any $r_0 > 0$ and $\beta \in (0,1)$, with probability at least $1-\beta$, the inequality*

$$\sup_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)} \left\{D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right\} \leq 3\bar{r}r_0 B_x\sqrt{\frac{4d + \ln(1/\beta)}{n}} + \lambda r_0^2,$$

holds, where $D_h(\boldsymbol{\delta}, \boldsymbol{\theta}) = \mathcal{L}_h(\boldsymbol{\theta} + \boldsymbol{\delta}) - \mathcal{L}_h(\boldsymbol{\theta})$ and $\widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}) = \widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta} + \boldsymbol{\delta}) - \widehat{\mathcal{L}}_h^{\#}(\boldsymbol{\theta})$ are same functions as in the proof for Lemma 4.4, the set $\partial\mathcal{B}(r_0) = \left\{\boldsymbol{\delta} \in \mathbb{R}^d \middle| \|\boldsymbol{\delta}\|_2 = r_0\right\}$, and $\bar{r} = \max\{1-r, r\}$.

Proof. (*Proof of Lemma B.1*) For any given radius $r_0 > 0$, denote $\alpha_\epsilon(r_0) = \frac{n(1-\epsilon)}{2\bar{r}r_0}$, and define a new random variable

$$\Delta_\epsilon(r_0) := \alpha_\epsilon(r_0) \sup_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)} \left\{D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h^{\#}(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right\},$$

parameterized by a constant $\epsilon \in (0,1)$ to be determined later. To investigate the tail performance of $\Delta_\epsilon(r_0)$, we apply Chernoff bound and obtain

$$\mathbb{P}\left[\Delta_\epsilon(r_0) \geq u\right] \leq \inf_{k>0}\left\{\exp\left(\ln^{\mathbb{E}_{\mathcal{D}}\left[e^{k\Delta_\epsilon(r_0)}\right]} - ku\right)\right\}, \ \forall u > 0. \tag{20}$$

Hence, we can first control the moment generating function (MGF) of $\Delta_\epsilon(r_0)$. Define $g(\boldsymbol{\delta}, \boldsymbol{\theta}^*) = \|\boldsymbol{\theta}^* + \boldsymbol{\delta}\|_2^2 - \|\boldsymbol{\theta}^*\|_2^2$, then the moment generating function is, for any $k > 0$,

$$\begin{aligned}
\mathbb{E}_{\mathcal{D}}\left[e^{k\Delta_\epsilon(r_0)}\right] &= \mathbb{E}_{\mathcal{D}}\left[\exp\left(\frac{kn(1-\epsilon)}{2\bar{r}r_0}\sup_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)}\left\{D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \lambda g(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right\}\right)\right] \\
&\leq \mathbb{E}_{\mathcal{D}}\left[\exp\left(\frac{kn(1-\epsilon)}{2\bar{r}r_0}\sup_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)}\left\{D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right\} - k\lambda\alpha_\epsilon(r_0)\inf_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)} g(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right)\right] \\
&= \mathbb{E}_{\mathcal{D}}\left[\exp\left(\frac{kn(1-\epsilon)}{2\bar{r}r_0}\sup_{\boldsymbol{\delta} \in \partial\mathcal{B}(r_0)}\left\{D_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*) - \widehat{D}_h(\boldsymbol{\delta}, \boldsymbol{\theta}^*)\right\}\right)\right] \cdot \exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right).
\end{aligned} \tag{21}$$

The last line comes from the fact

$$\inf_{\boldsymbol{\delta}\in\partial\mathcal{B}(r_0)} g\left(\boldsymbol{\delta},\boldsymbol{\theta}^*\right) = \inf_{\boldsymbol{\delta}\in\partial\mathcal{B}(r_0)} \|\boldsymbol{\delta}\|_2^2 + 2\boldsymbol{\delta}^\top\boldsymbol{\theta}^* = -r_0^2 \quad =: -\beta(r_0) < 0.$$

By Rademacher Symmetrization Lemma (see, for example, Proposition 4.11 in (Wainwright, 2019)) and noticing that the exponential function is convex and increasing, we can further upper bound (21) with Rademacher Complexity as shown below,

$$(21) \leq \mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(\frac{kn(1-\epsilon)}{\bar{r}r_0}\sup_{\boldsymbol{\delta}\in\partial\mathcal{B}(r_0)}\left\{\frac{1}{n}\sum_{i=1}^n z_i d_h(\boldsymbol{\delta},\boldsymbol{\theta}^*;\boldsymbol{x}_i,y_i)\right\}\right)\right]\cdot\exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right), \tag{22}$$

where $z_i, \forall i \in \{1,2,\ldots,n\}$ are independent Rademacher random variables. Note that the function $d_h(\boldsymbol{\delta},\boldsymbol{\theta}^*;\boldsymbol{x}_i,y_i) = \ell_h\left(\boldsymbol{\delta}+\boldsymbol{\theta}^*;\boldsymbol{x}_i,y_i\right) - \ell_h\left(\boldsymbol{\theta}^*;\boldsymbol{x}_i,y_i\right) = c_h\left(\epsilon_i - \boldsymbol{\delta}^\top\boldsymbol{x}_i\right) - c_h\left(\epsilon_i\right)$ is $\bar{r}$-lipschitz continuous in $\langle\boldsymbol{\delta},\boldsymbol{x}_i\rangle$, and $d_h(\cdot)=0$ when $\langle\boldsymbol{\delta},\boldsymbol{x}_i\rangle = 0$. Therefore, by Ledoux-Talagrand contraction inequality, we have

$$\begin{aligned}(22) &\leq \mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(\frac{kn(1-\epsilon)}{r_0}\sup_{\boldsymbol{\delta}\in\partial\mathcal{B}(r_0)}\left\{\frac{1}{n}\sum_{i=1}^n z_i\left\langle\boldsymbol{\delta},\boldsymbol{x}_i\right\rangle\right\}\right)\right]\cdot\exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right)\\ &= \mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(k(1-\epsilon)\left\|\sum_{i=1}^n z_i\boldsymbol{x}_i\right\|_2\right)\right]\cdot\exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right),\end{aligned} \tag{23}$$

where the second line is obtained by cancelling out n and then plugging in the maximizer $\boldsymbol{\delta}^* = r_0\cdot\frac{\sum_{i=1}^n z_i\boldsymbol{x}_i}{\left\|\sum_{i=1}^n z_i\boldsymbol{x_i}\right\|_2}$.

Denote $\boldsymbol{w} := \sum_{i=1}^n z_i\boldsymbol{x}_i$. Now, we focus on the ℓ_2-norm term $\|\boldsymbol{w}\|_2$. Let a set of d-dimensional points $\{\boldsymbol{p}_1,\boldsymbol{p}_2,\ldots,\boldsymbol{p}_{N_\epsilon}\}$ with cardinality N_ϵ be the *ϵ-covering* of a unit ball $\mathcal{B}$ with respect to ℓ_2 norm. Then, we know that there must exist a point $\boldsymbol{p}_j\in\mathcal{B}, j\in\{1,2,\ldots,N_\epsilon\}$ such that

$$(1-\epsilon)\|\boldsymbol{w}\|_2 \leq \langle\boldsymbol{p}_j,\boldsymbol{w}\rangle;$$

otherwise, the unit ball $\mathcal{B}$ is not covered by the given *ϵ-covering* $\{\boldsymbol{p}_1,\boldsymbol{p}_2,\ldots,\boldsymbol{p}_{N_\epsilon}\}$. Therefore, we are able to further upper bound (23) with

$$\begin{aligned}(23) &\leq \mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(k\cdot\max_{j\in\{1,\ldots,N_\epsilon\}}\langle\boldsymbol{p}_j,\boldsymbol{w}\rangle\right)\right]\cdot\exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right)\\ &\leq \exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right)\cdot\sum_{j=1}^{N_\epsilon}\mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(k\cdot\langle\boldsymbol{p}_j,\boldsymbol{w}\rangle\right)\right]\end{aligned} \tag{24}$$

For any given $\boldsymbol{p}_j\in\mathcal{B}$, we note that $\mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(k\cdot\langle\boldsymbol{p}_j,\boldsymbol{w}\rangle\right)\right] = \mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(k\sum_{i=1}^n v_{i,j}\right)\right]$ is the MGF of the sum over n independent zero-mean random variable $v_{i,j} := \langle\boldsymbol{p}_j, z_i\boldsymbol{x}_i\rangle, \forall i=1,\ldots,n$. Moreover, since we assume $\boldsymbol{x}\in\mathcal{B}(B_x)$, the random variable $v_{i,j}$ is bounded almost surely as $|v_{i,j}|\leq B_x$. Thus, $v_{i,j}$ belongs to $\mathsf{SubGaussian}(\sigma^2)$ with $\sigma^2 = B_x^2$, and

$$\mathbb{E}_{\mathcal{D},z_i}\left[\exp\left(k\cdot\langle\boldsymbol{p}_j,\boldsymbol{w}\rangle\right)\right] \leq \exp\left(\frac{nk^2\sigma^2}{2}\right),\ \forall\boldsymbol{p}_j\in\mathcal{B},\forall k>0, \tag{25}$$

Plugging (25) into (24), and combining (21), (22), (23), and (24) all together, we finally obtain an upper bound for the MGF of $\Delta_\epsilon(r_0)$ as below,

$$\mathbb{E}_{\mathcal{D}}\left[e^{k\Delta_\epsilon(r_0)}\right] \leq \exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0)\right)\cdot\sum_{j=1}^{N_\epsilon}\exp\left(\frac{nk^2\sigma^2}{2}\right), \forall\epsilon\in(0,1),\forall r_0>0,\forall k>0, \tag{26}$$

with $\alpha_\epsilon(r_0) = \frac{n(1-\epsilon)}{2\bar{r}r_0}$ and $\beta(r_0) = r_0^2$.

Replacing the MGF of $\Delta_\epsilon(r_0)$ in (20) with its upper bound (26), we obtain that, for any $u>0$,

$$\begin{aligned}\mathbb{P}\left[\Delta_\epsilon(r_0)\geq u\right] &\leq \inf_{k>0}\left\{\exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0) + \ln^{N_\epsilon} + \frac{nk^2\sigma^2}{2} - ku\right)\right\}\\ &\leq \inf_{k>0}\left\{\exp\left(k\lambda\alpha_\epsilon(r_0)\beta(r_0) + d\ln\left(1+\frac{2}{\epsilon}\right) + \frac{nk^2\sigma^2}{2} - ku\right)\right\}\end{aligned} \tag{27}$$

where the second inequality is by the fact that the covering number N_ϵ is known to satisfy $N_\epsilon \le (1+\frac{2}{\epsilon})^d$. After replace u with $\alpha_\epsilon(r_0)\cdot\left(\frac{2\sqrt{2}\bar{r}r_0\sigma}{1-\epsilon}\sqrt{\frac{v}{n}}+\lambda\beta(r_0)\right)$ for any $v>0$, (27) becomes, for any $v>0$,

$$\begin{aligned}
&\mathbb{P}\left[\sup_{\boldsymbol{\delta}\in\mathbb{B}(r_0)}\left\{D_h(\boldsymbol{\delta},\boldsymbol{\theta}^*)-\widehat{D}_h^{\#}(\boldsymbol{\delta},\boldsymbol{\theta}^*)\right\}\ge\frac{2\sqrt{2}\bar{r}r_0\sigma}{1-\epsilon}\sqrt{\frac{v}{n}}+\lambda\beta(r_0)\right]\\
&\le\inf_{k>0}\left\{\exp\left(d\ln\left(1+\frac{2}{\epsilon}\right)+\frac{nk^2\sigma^2}{2}-k\sigma\sqrt{2vn}\right)\right\}\\
&=\exp\left(d\ln\left(1+\frac{2}{\epsilon}\right)-v\right),
\end{aligned}\tag{28}$$

where the last line is by taking $k^*=\sigma\sqrt{\frac{2v}{n}}$. Lastly, if we set $\epsilon=\frac{2}{e^4-1}$, then $\frac{2\sqrt{2}}{1-\epsilon}\le 3$ and the right-hand-side of (28) becomes $\exp(4d-v)$. Set $\exp(4d-v)=\beta$ and solve for v, we get the desired lemma. □

B.4. Proof of Theorem 4.5

Proof. This Theorem is a direct conclusion from combing Lemma 4.2, 4.3, and 4.4. Specifically, if we set $\lambda=\Theta\left(\sqrt{\frac{1}{n\varepsilon}+\frac{d\sigma^2}{n^2}}\right)$ and $h=\Theta(1/\sqrt{n\varepsilon\lambda})$ as same values in Theorem 3.6, then with probability at least $1-\beta,\forall\beta\in(0,1)$ over random draw of samples, we have (by Lemma 4.4),

$$\left\|\widehat{\boldsymbol{\theta}}_h^{\#}-\boldsymbol{\theta}_h^*\right\|_2\le O\left(\frac{l_1\kappa_1}{\rho_1\underline{f}n\varepsilon}+\frac{L}{\rho_1\underline{f}}\sqrt{\frac{d+\ln(1/\beta)}{n}}\right).$$

Since we assume that $\varepsilon=\omega(1/\sqrt{n})$, naturally we have $h^2=o(1/\sqrt{n})$. Similarly, replacing h^2, we have $\|\boldsymbol{\theta}_h^*-\boldsymbol{\theta}^*\|_2\le\frac{h^2l_1\kappa_2}{\rho_1(\underline{f}-hl_1\kappa_2)}\le O\left(\frac{l_1\kappa_2}{\rho_1\underline{f}n\varepsilon}\right)$. Putting three Lemmas together completes the proof:

$$\mathbb{E}_{\mathsf{OP}}\left[\left\|\widehat{\boldsymbol{\theta}}_h^{\mathsf{OP}}(\boldsymbol{b})-\boldsymbol{\theta}^*\right\|_2\right]\le O\left(L\sqrt{\frac{d\ln(1/\delta)}{n\varepsilon}}+\frac{L}{\rho_1\underline{f}}\sqrt{\frac{d+\ln(1/\beta)}{n}}+\frac{l_1\kappa_2}{\rho_1\underline{f}n\varepsilon}\right)=O\left(\max\left\{\sqrt{\frac{d}{n}},\sqrt{\frac{d\ln(1/\delta)}{n\varepsilon}}\right\}\right).$$

□

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